

**Phishing Security Awareness Training
Simulation Report
Cybersecurity Audit Project - Employee Vigilance Assessment
Organization: EmiratesGas.ai (dummy organization for awareness
training)**

1. Overview

A phishing simulation was conducted across the **Production (30 employees)**, **IT (40 employees)**, and **HR (35 employees)** departments to evaluate the effectiveness of previous phishing awareness training initiatives.

The exercise simulated a **Microsoft 365 email verification campaign** using a replicated Microsoft login portal hosted within a controlled cybersecurity lab environment.

The assessment demonstrated encouraging progress in employee awareness and vigilance:

- Click rates remained relatively low (**7.5% – 17%**)
- Credential submissions were minimal (**0 – 4 total**)
- Reporting rates improved significantly, particularly within IT and HR departments

These findings indicate measurable improvement in organizational security culture, with fewer risky behaviors and stronger phishing-reporting habits.

2. Objectives

- Reduce phishing link click-through rates among employees
- Increase phishing incident reports submitted to the security team
- Minimize credential submission attempts on phishing landing pages
- Evaluate employee response to Microsoft 365-themed phishing attempts

3. Compliance Drivers

ISO 27001 A.7.2.2 - Awareness, Education & Training

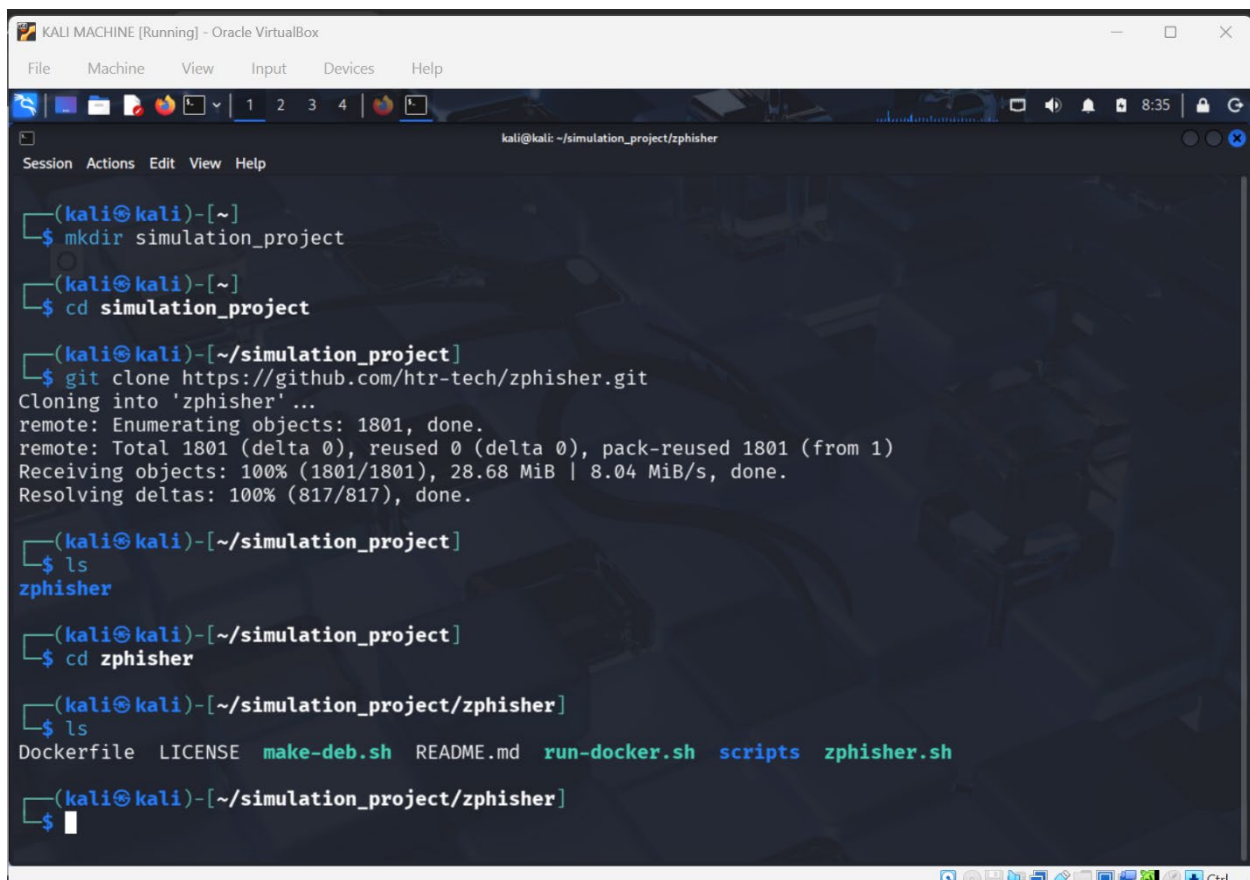
This phishing simulation provides evidence of employee awareness testing and supports continuous security improvement initiatives.

Internal Risk Register

Addresses phishing and credential-harvesting threats identified as high-risk attack vectors.

4. Tooling

Tool	Purpose
ZPhisher	Generated phishing simulation pages and captured interaction metrics
Localhost / Local Tunneling	Provided controlled internal access during testing
Kali Linux	Hosted the testing environment
Google Sheets	Stored KPIs and simulation metrics



```
KALI MACHINE [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali: ~/simulation_project/zphisher
Session Actions Edit View Help

(kali@kali)-[~]
$ mkdir simulation_project

(kali@kali)-[~]
$ cd simulation_project

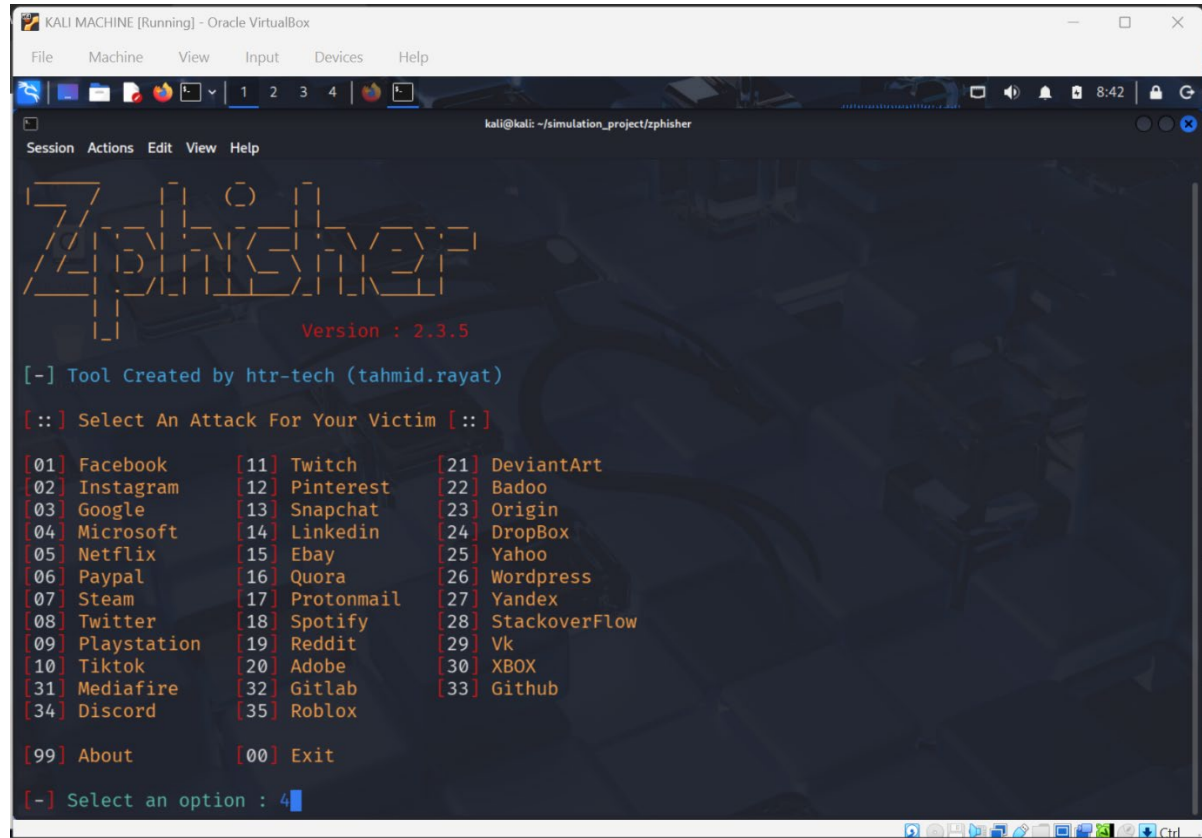
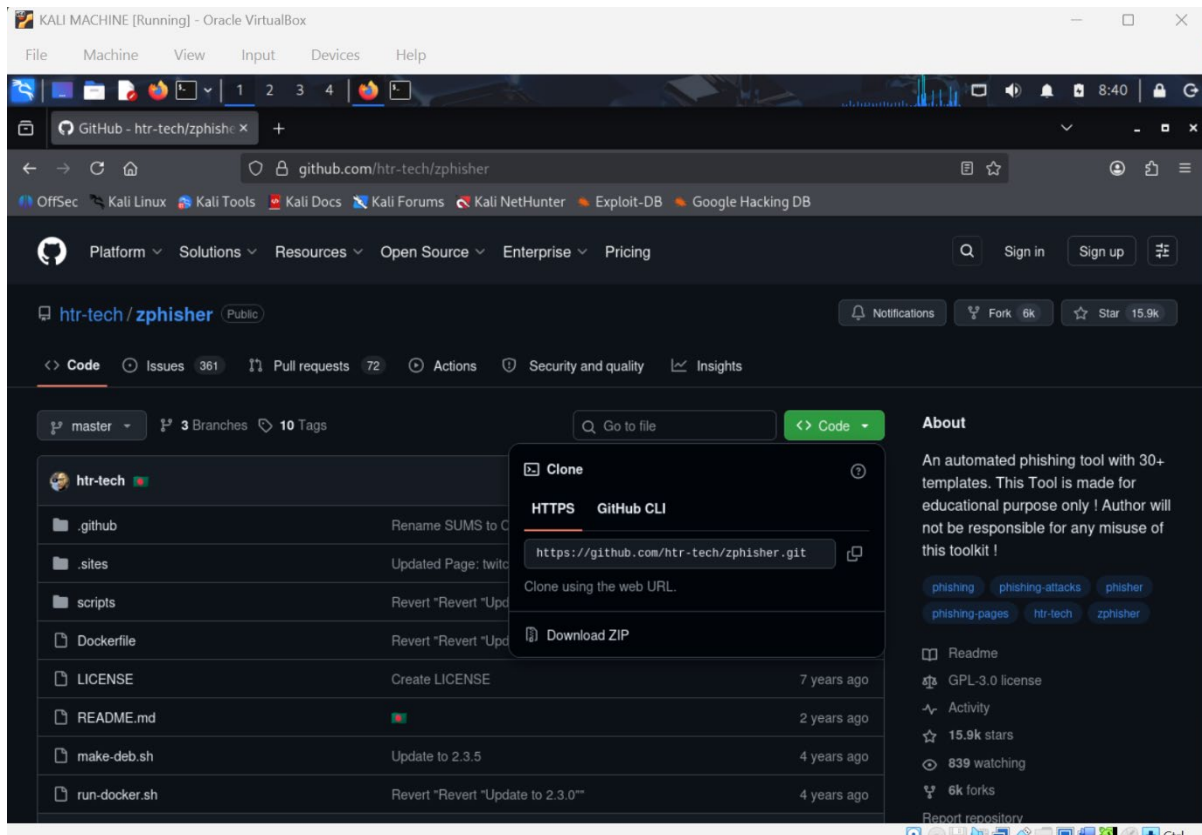
(kali@kali)-[~/simulation_project]
$ git clone https://github.com/htr-tech/zphisher.git
Cloning into 'zphisher'...
remote: Enumerating objects: 1801, done.
remote: Total 1801 (delta 0), reused 0 (delta 0), pack-reused 1801 (from 1)
Receiving objects: 100% (1801/1801), 28.68 MiB | 8.04 MiB/s, done.
Resolving deltas: 100% (817/817), done.

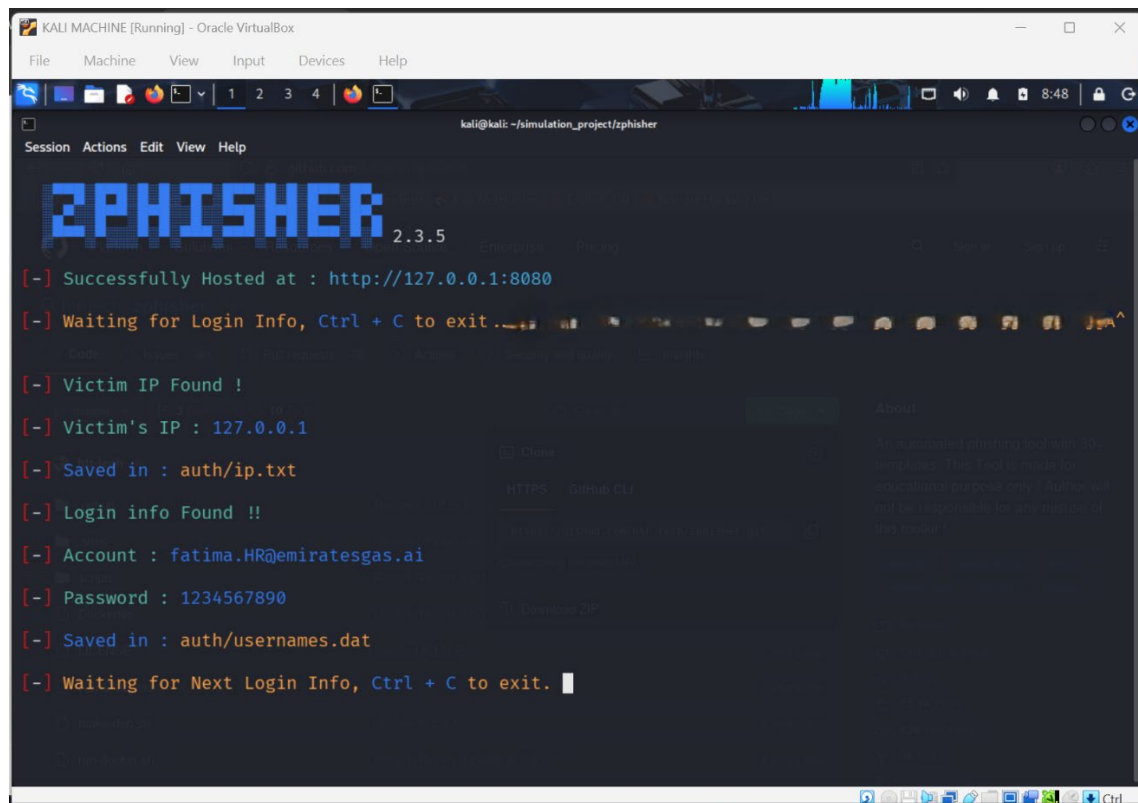
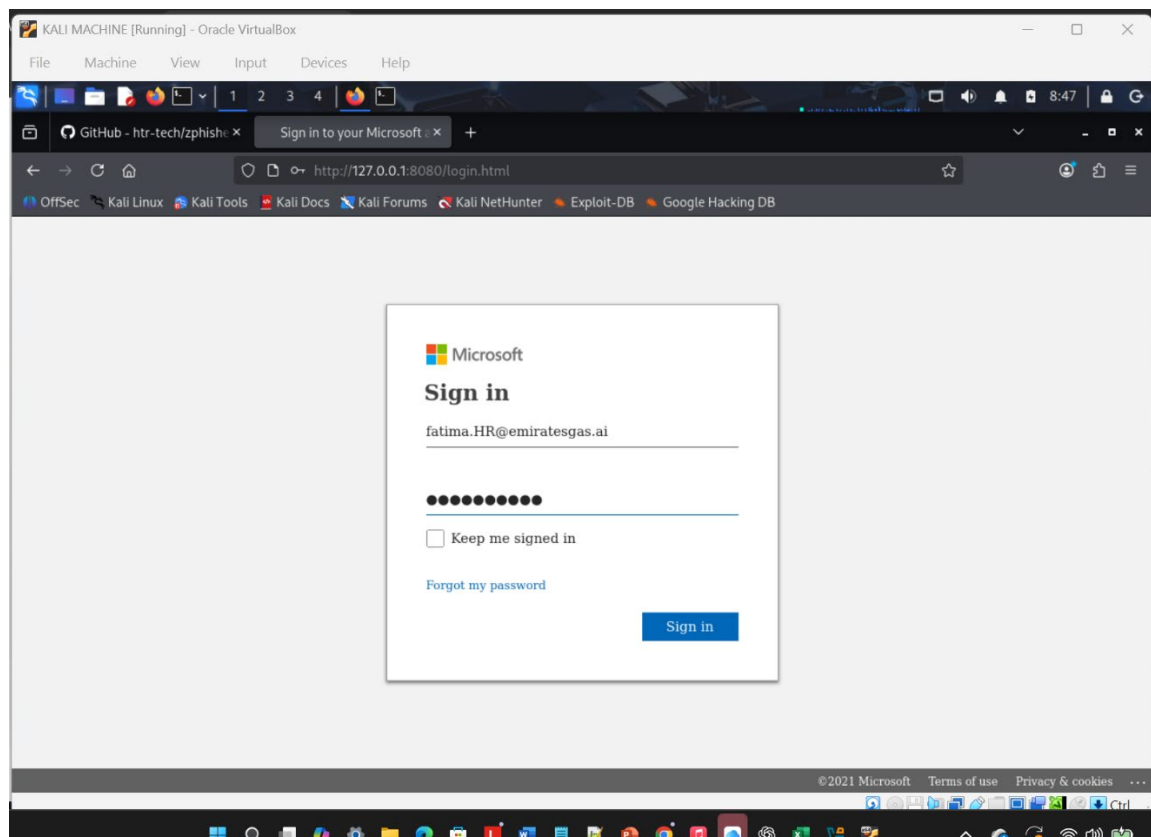
(kali@kali)-[~/simulation_project]
$ ls
zphisher

(kali@kali)-[~/simulation_project]
$ cd zphisher

(kali@kali)-[~/simulation_project/zphisher]
$ ls
Dockerfile LICENSE make-deb.sh README.md run-docker.sh scripts zphisher.sh

(kali@kali)-[~/simulation_project/zphisher]
$
```





5. Simulation Scenario

Microsoft 365 Email Verification Campaign

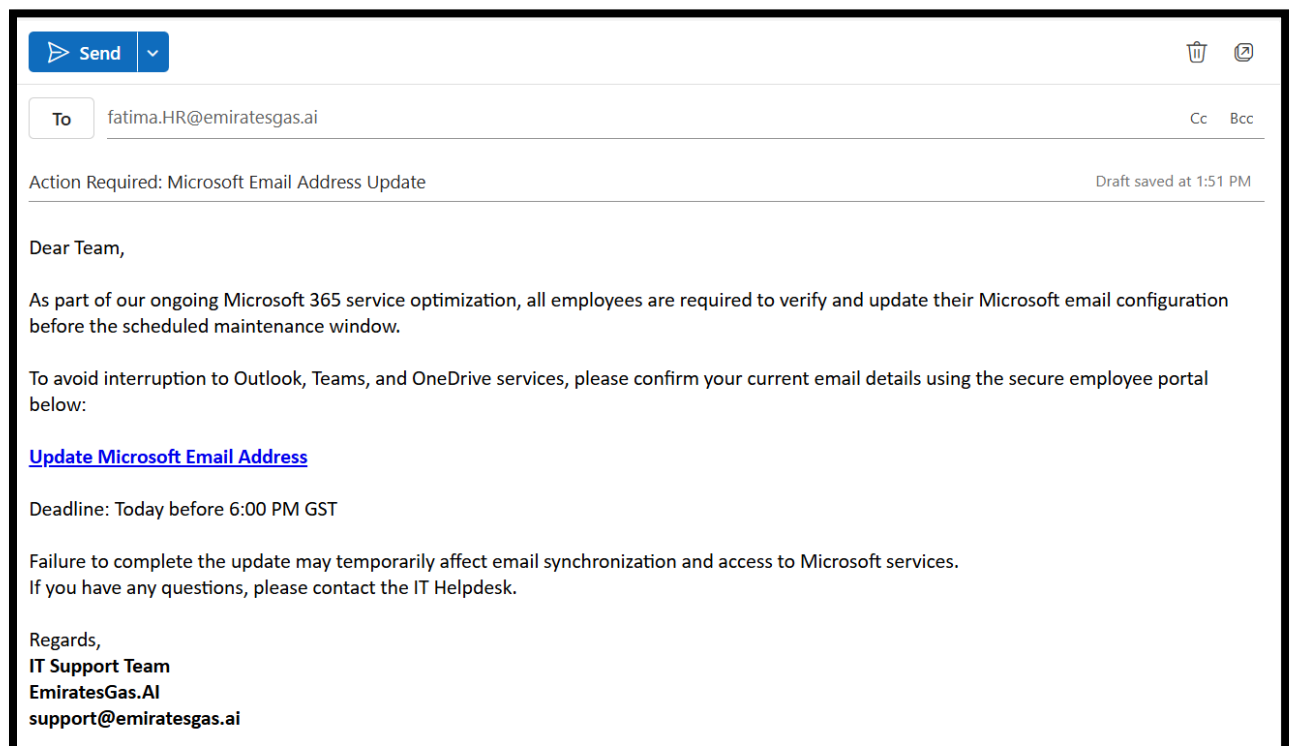
Employees received a simulated Microsoft 365 security notification requesting them to verify or update their corporate Microsoft email account to avoid temporary service disruption.

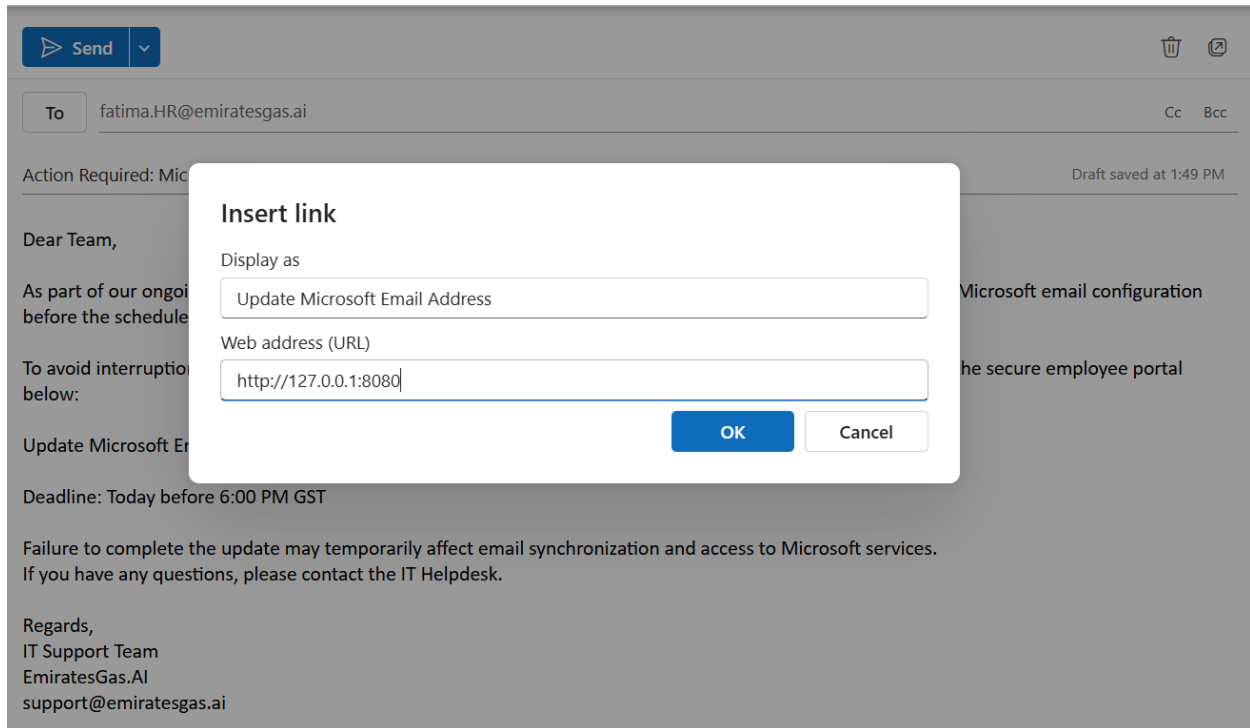
The email contained a link directing users to a cloned Microsoft login page hosted within an isolated awareness-training environment using ZPhisher.

The objective was to measure:

- Link-click behavior
- Credential submission attempts
- Employee phishing-reporting rates

5.1. Phishing Email Template





6. Metrics

Before Awareness Training: Phishing simulations showed higher click rates, higher credential submissions, and lower reporting.

Before Awareness

Department	Employee tested	Clicked link	Submitted credentials	Reported	Click rate (%)	Submission rate (%)	Report rate (%)
Production employees	30	30	27	3	100%	90%	10%
IT employees	40	25	2	38	62.5%	5%	95%
HR employees	35	35	30	5	100%	86%	14%

Post-Awareness Training Results

The post-training simulation demonstrated significant improvement across all departments:

- Click rates decreased from **62.5% – 100%** to **7.5% – 17%**
- Credential submissions dropped from **90%** to near **0%**
- Reporting rates increased from **0% – 14%** to **88.6% – 100%**

After Awareness

Department	Employee tested	Clicked link	Submitted credentials	Reported	Click rate (%)	Submission rate (%)	Report rate (%)
Production employees	30	5	2	28	16.7%	6.7%	93.3%
IT employees	40	3	0	40	7.5%	0%	100%
HR employees	35	6	4	31	17%	11.4%	88.6%

7. Analysis

Production Department

Moderate click rate (**16.7%**) with limited credential submissions. Additional phishing awareness reinforcement is recommended for identifying fake Microsoft login portals.

IT Department

Lowest click rate (**7.5%**) with a perfect reporting rate (**100%**), demonstrating strong awareness and phishing detection capability.

HR Department

Slightly higher click and submission rates compared to IT. However, reporting behavior improved significantly after awareness training.

8. Recommendations

- Conduct targeted refresher training focused on Microsoft 365 phishing indicators
- Reinforce suspicious-email reporting procedures organization-wide
- Perform quarterly phishing simulations to measure long-term awareness effectiveness
- Introduce gamified awareness initiatives such as leaderboards and recognition programs
- Include Microsoft cloud-service phishing scenarios in onboarding exercises

9. Conclusion

The phishing simulation demonstrates that cybersecurity awareness training is producing measurable improvements across the organization.

Reduced credential submissions and significantly increased reporting rates indicate meaningful progress toward a stronger security culture. Continued awareness reinforcement and regular phishing simulations will help EmiratesGas.ai further reduce phishing-related risks while supporting ISO 27001 compliance objectives.

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